ines certify

mopo

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Chapter 1

Proof of concept for the certification process

As the ines spec describes an interoperable data format, it is particularly interesting to define a certification process in that format such that it can be used to certify different energy system models for the correct implementation of certain features such as, e.g., different time resolutions in different parts of the model.

Here, we only consider a proof of concept to show the potential of certification with the ines spec. Once a consortium has been formed around the ines spec, it makes sense to continue the development of the certification process. In the following, we cherry pick some setups that show different aspects of energy system models. Some suggestions for expansions on these setups can be found in appendix A.

While it is possible to describe different features in the ines spec, there is not actually a specific model associated to it. Therefore, if we want to obtain the exact solution for the certification process, we'll have to solve the system manually. In the following, not only do we provide a description of the setup but also provide the specific equations to solve that setup.

Figure 1.1 shows the workflow in Spine Toolbox and Table 1.1 gives a brief overview of the setups.

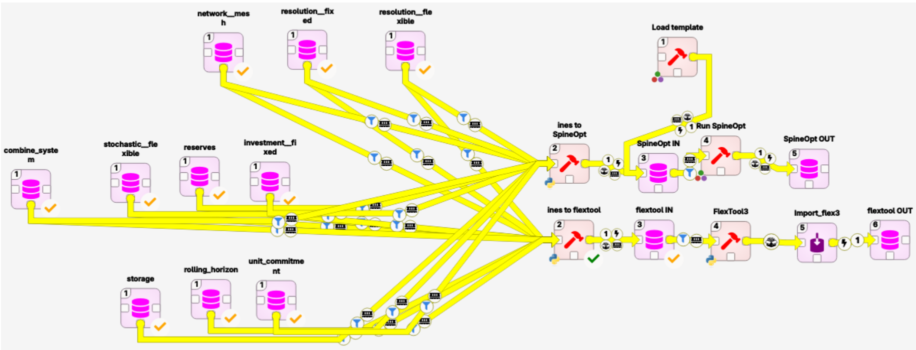

Figure 1.1: Certification workflow

name	description
network = meshed	Energy balance where the demand equals the supply for a single time step. Triangular (but one directional) network with the two supply units and demand unit. The cheapest unit should be chosen.
resolution = fixed	network = meshed but expanded to a time series. Both units have the same resolution.
resolution = flexible	resolution = fixed but the two units have different resolutions (1h for expensive unit and 2h for cheap unit).
stochastic	resolution = fixed but there are scenarios for the demand and the resource prices.
reserves	resolution = fixed but there is a demand for reserves. The expensive unit is able to provide reserves but not enough to entirely avoid load shedding.
unit commitment	resolution = fixed but there is unit commitment for both units. There is a minimum up time, a minimum down time and a part load efficiency.
investment = fixed	resolution = fixed but there are investment variables.
storage	resolution = fixed but with storage and renewable unit.
rolling horizon	investment = fixed but there are two horizons and the model rolls from one horizon to the other. The demand is increased in the second horizon.
combined system	A combination of all the previous setups. Sector coupled for the flexible time steps. The gas sector consists of import, demand and gas provision to a gas plant. The electricity sector consists of the gas plant, a nuclear plant, batteries, onshore PV, offshore wind, electricity import and demand. The demand is in the distribution grid, the import and offshore wind is in a separate node. There are stochastics for the renewables and import prices.

Table 1.1: Expansion of the certification process.

1.1 network = mesh

The goal of the setup is to test the energy balance in space. To that end the demand and supply are scattered over different nodes. They are connected to each other through directional lines with small losses but sufficient capacity. The lines are directional to avoid circular flows and the lines have losses to avoid multiple possible solutions. One of the supply units is obviously cheaper such that it is very clear what the solution should be.

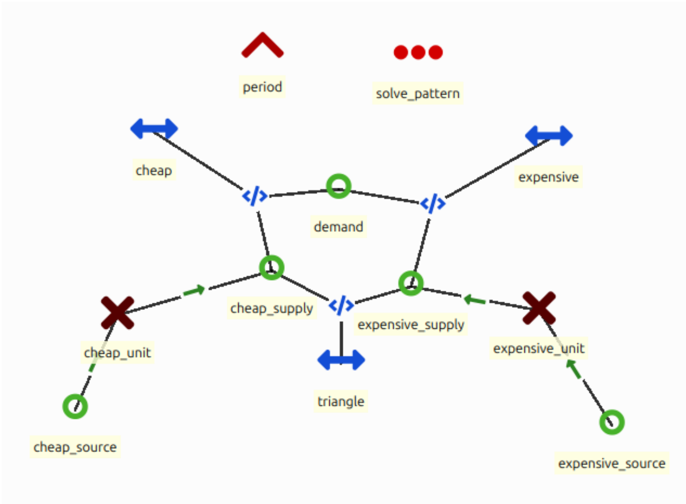

Figure 1.2: Implementation of network = mesh in the ines spec.

Formulation to obtain the exact solution for this setup:

minimise:

$$z = p^{other\_operational\_costs}_{cheap\_source} \cdot v^{flow}_{cheap\_source} + p^{other\_operational\_costs}_{expensive\_source} \cdot v^{flow}_{expensive\_source}$$

subject to:

$$\begin{aligned}
p^{flow_profile}_{demand} &= v^{flow}_{cheap_link,out} + v^{flow}_{expensive_link,out} \\
v^{flow}_{cheap_link,out} &= p^{efficiency}_{cheap_link} \cdot v^{flow}_{cheap_link,in} \\
v^{flow}_{expensive_link,out} &= p^{efficiency}_{expensive_link} \cdot v^{flow}_{expensive_link,in} \\
v^{flow}_{triangle,out} &= p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in} \\
v^{flow}_{cheap_link,in} &= v^{flow}_{cheap_supply} + v^{flow}_{triangle,out} \\
v^{flow}_{expensive_supply} &= v^{flow}_{expensive_link,in} + v^{flow}_{triangle,in} \\
v^{flow}_{expensive_supply} &= p^{efficiency}_{expensive_unit} \cdot v^{flow}_{expensive_source} \\
v^{flow}_{cheap_supply} &= p^{efficiency}_{cheap_unit} \cdot v^{flow}_{cheap_source}
\end{aligned}$$

bound to:

$$
\begin{aligned}
0 \leq & v^{flow}_{cheap\_link,in} \\
0 \leq & v^{flow}_{cheap\_link,out} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq & v^{flow}_{expensive\_link,in} \\
0 \leq & v^{flow}_{expensive\_link,out} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq & v^{flow}_{triangle,in} \\
0 \leq & v^{flow}_{triangle,out} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle} \\
0 \leq & v^{flow}_{expensive\_supply} && \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
0 \leq & v^{flow}_{expensive\_source} \\
0 \leq & v^{flow}_{cheap\_supply} && \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \\
0 \leq & v^{flow}_{cheap\_source}
\end{aligned}
$$

1.2 resolution = fixed

The goal of the setup is to test the energy balance in time. To that end, the setup starts from network = meshed and adds a time series for the demand and the supply. All entities have the same resolution.

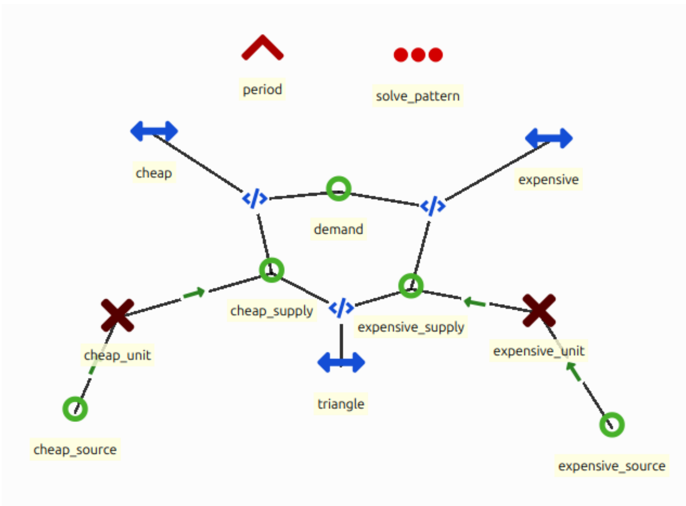

Figure 1.3: Implementation of resolution = fixed in the ines spec.

Formulation to obtain the exact solution for this setup:

minimise:

$$
z = \sum_t p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} + p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t}
$$

subject to:

$$
\begin{aligned}
p^{flow\_profile}_{demand,t} =& v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t} \\
v^{flow}_{cheap\_link,out,t} =& p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t} \\
v^{flow}_{expensive\_link,out,t} =& p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t} \\
v^{flow}_{triangle,out,t} =& p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t} \\
v^{flow}_{cheap\_link,in,t} =& v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t} \\
v^{flow}_{expensive\_supply,t} =& v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t} \\
v^{flow}_{expensive\_supply,t} =& p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t} \\
v^{flow}_{cheap\_supply,t} =& p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
0 \leq& v^{flow}_{cheap\_link,in,t} \\
0 \leq& v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq& v^{flow}_{expensive\_link,in,t} \\
0 \leq& v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq& v^{flow}_{triangle,in,t} \\
0 \leq& v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle} \\
0 \leq& v^{flow}_{expensive\_supply,t} && \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
0 \leq& v^{flow}_{expensive\_source,t} \\
0 \leq& v^{flow}_{cheap\_supply,t} && \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \\
0 \leq& v^{flow}_{cheap\_source,t}
\end{aligned}
$$

1.3 resolution = flexible

The goal of this setup is to test the energy balance in time and space. To that end, the setup starts from resolution = fixed but reduces the resolution of one of the supply units. The resolution is only changed at the source such that the supply of the unit is still able to provide the same output as before.

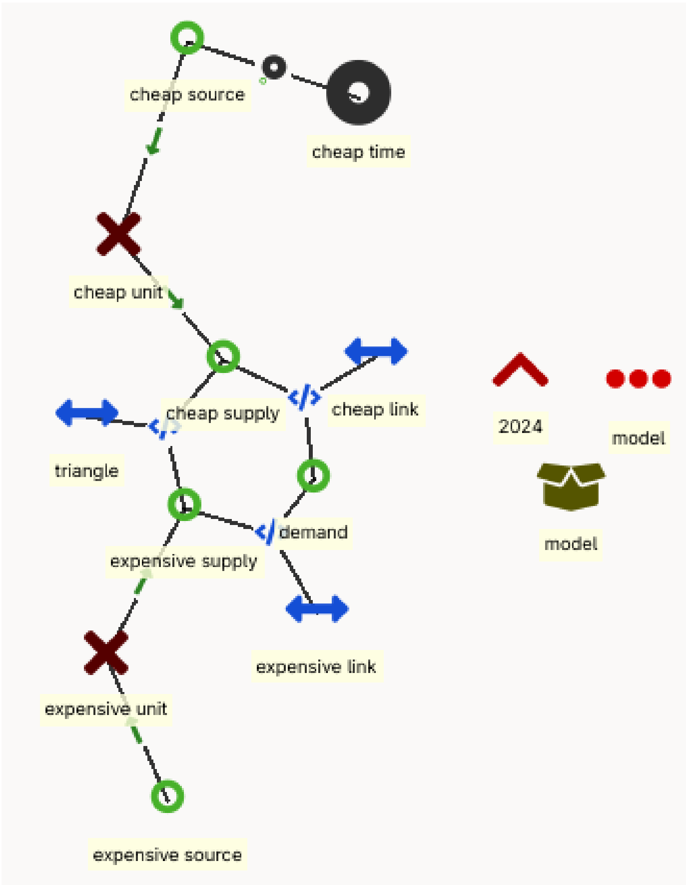

Figure 1.4: Implementation of resolution = flexible in the ines spec.

Formulation to obtain the exact solution for this setup:

minimise:

$$z = \sum_{t \in t_{cheap}} p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} + \sum_{t} p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t}$$

subject to:

$$
\begin{aligned}
p^{flow\_profile}_{demand,t} =& v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t} \\
v^{flow}_{cheap\_link,out,t} =& p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t} \\
v^{flow}_{expensive\_link,out,t} =& p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t} \\
v^{flow}_{triangle,out,t} =& p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t} \\
v^{flow}_{cheap\_link,in,t} =& v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t} \\
v^{flow}_{expensive\_supply,t} =& v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t} \\
v^{flow}_{expensive\_supply,t} =& p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t} \\
\sum_{t \in t_{cheap}} v^{flow}_{cheap\_supply,t} \cdot 1h =& p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t_{cheap}} \cdot \Delta_{t_{cheap}}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
0 \leq& v^{flow}_{cheap\_link,in,t} & \\
0 \leq& v^{flow}_{cheap\_link,out,t} & \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq& v^{flow}_{expensive\_link,in,t} & \\
0 \leq& v^{flow}_{expensive\_link,out,t} & \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq& v^{flow}_{triangle,in,t} & \\
0 \leq& v^{flow}_{triangle,out,t} & \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle} \\
0 \leq& v^{flow}_{expensive\_supply,t} & \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
0 \leq& v^{flow}_{expensive\_source,t} & \\
0 \leq& v^{flow}_{cheap\_supply,t} & \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \\
0 \leq& v^{flow}_{cheap\_source,t} & \leq \frac{1}{p^{efficiency}_{cheap\_unit}} \cdot p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit}
\end{aligned}
$$

1.4 stochastic

The goal of this setup is to test scenarios in the formulation. To that end the setup starts from resolution = fixed and adds stochastics for the demand and the commodity prices. The demand can be high and low and the commodity prices can be expensive or cheap. Combining these scenarios, results in 4 equally likely scenarios for the entire model.

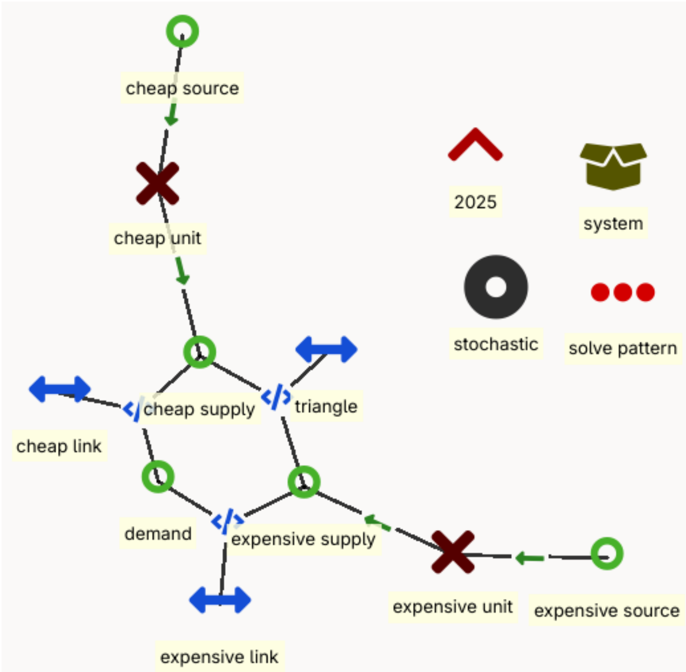

Figure 1.5: stochastic = flexible

Formulation to obtain the exact solution for this setup:

minimise:

$$z = \sum_s p_s^{weight} \cdot \sum_t p_{cheap\_source,t,s}^{other\_operational\_costs} \cdot v_{cheap\_source,t,s}^{flow} + p_{expensive\_source,t,s}^{other\_operational\_costs} \cdot v_{expensive\_source,t,s}^{flow}$$

subject to:

$$\begin{aligned}
p_{demand,t,s}^{flow_profile} =& v_{cheap_link,out,t,s}^{flow} + v_{expensive_link,out,t,s}^{flow} \\
v_{cheap_link,out,t,s}^{flow} =& p_{cheap_link}^{efficiency} \cdot v_{cheap_link,in,t,s}^{flow} \\
v_{expensive_link,out,t,s}^{flow} =& p_{expensive_link}^{efficiency} \cdot v_{expensive_link,in,t,s}^{flow} \\
v_{triangle,out,t,s}^{flow} =& p_{triangle}^{efficiency} \cdot v_{triangle,in,t,s}^{flow} \\
v_{cheap_link,in,t,s}^{flow} =& v_{cheap_supply,t,s}^{flow} + v_{triangle,out,t,s}^{flow} \\
v_{expensive_supply,t,s}^{flow} =& v_{expensive_link,in,t,s}^{flow} + v_{triangle,in,t,s}^{flow} \\
v_{expensive_supply,t,s}^{flow} =& p_{expensive_unit}^{efficiency} \cdot v_{expensive_source,t,s}^{flow} \\
v_{cheap_supply,t,s}^{flow} =& p_{cheap_unit}^{efficiency} \cdot v_{cheap_source,t,s}^{flow}
\end{aligned}$$

bound to:

$$0 \leq v^{flow}_{cheap\_link,in,t,s}$$
$$0 \leq v^{flow}_{cheap\_link,out,t,s} \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link}$$
$$0 \leq v^{flow}_{expensive\_link,in,t,s}$$
$$0 \leq v^{flow}_{expensive\_link,out,t,s} \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link}$$
$$0 \leq v^{flow}_{triangle,in,t,s}$$
$$0 \leq v^{flow}_{triangle,out,t,s} \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle}$$
$$0 \leq v^{flow}_{expensive\_supply,t,s} \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit}$$
$$0 \leq v^{flow}_{expensive\_source,t,s}$$
$$0 \leq v^{flow}_{cheap\_supply,t,s} \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit}$$
$$0 \leq v^{flow}_{cheap\_source,t,s}$$

1.5 reserves

The goal of this setup is to test the reserve requirements. To that end the setup starts from resolution = fixed and adds a reserve requirement to the demand. A common way to include reserves is by working with a reserve node, dismissing the capacity restrictions of the lines. For the proof of concept, we limit ourselves to upward reserves.

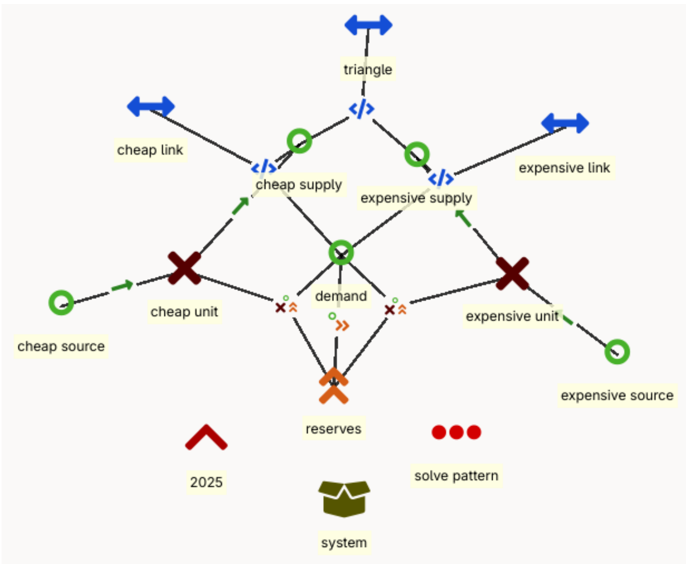

Figure 1.6: reserves

Formulation to obtain the exact solution for this setup:

minimise:

$$
\begin{aligned}
z = &\sum_t p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} + p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t} \\
&+ \sum_t p^{penalty}_{demand} \cdot v^{slack}_{demand,t} + p^{procurement\_cost}_{cheap\_supply,t} \cdot v^{flow,reserve}_{cheap\_supply,t} + p^{procurement\_cost}_{expensive\_supply,t} \cdot v^{flow,reserve}_{expensive\_supply,t}
\end{aligned}
$$

subject to:

$$
p^{reserve}_{demand,t} =< v^{flow,reserve}_{cheap\_supply,t} + v^{flow,reserve}_{expensive\_supply,t} + v^{slack}_{demand,t}
$$

$$
p^{flow\_profile}_{demand,t} = v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,out,t} &= p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t} \\
v^{flow}_{expensive\_link,out,t} &= p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t} \\
v^{flow}_{triangle,out,t} &= p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,in,t} &= v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t} \\
v^{flow}_{expensive\_supply,t} &= v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{expensive\_supply,t} &= p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t} \\
v^{flow}_{cheap\_supply,t} &= p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{expensive\_supply,t} + v^{flow,reserve}_{expensive\_supply,t} &\leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
v^{flow}_{cheap\_supply,t} + v^{flow,reserve}_{expensive\_supply,t} &\leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
0 &\leq v^{slack}_{demand,t} \\
0 &\leq v^{flow,reserve}_{cheap\_supply,t} && \leq p^{max\_reserve}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \\
0 &\leq v^{flow,reserve}_{expensive\_supply,t} && \leq p^{max\_reserve}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit}
\end{aligned}
$$

$$
\begin{aligned}
0 &\leq v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_links}_{cheap\_link} \\
0 &\leq v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_links}_{expensive\_link} \\
0 &\leq v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle}
\end{aligned}
$$

$$
\begin{aligned}
0 &\leq v^{flow}_{expensive\_supply,t} \\
0 &\leq v^{flow}_{cheap\_supply,t}
\end{aligned}
$$

1.6 unit commitment

The goal of this setup is to test the unit commitment functionality. To that end, the setup starts from resolution = fixed but adds on/off state variables to the supply units. The commitment is constrained by the minimum up and down time and there is a cost associated to starting units. The startup cost and minimum up and down time are large for the cheap unit to incentivise the model to choose the expensive unit.

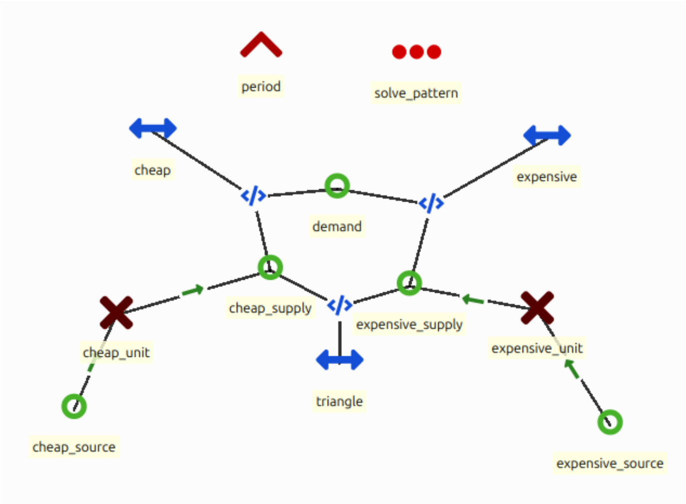

Figure 1.7: Implementation of unit commitment in the ines spec.

Formulation to obtain the exact solution for this setup:

minimise:

$$
\begin{aligned}
z = \sum_t & p^{startup\_cost}_{cheap\_unit} \cdot v^{on\_up.}_{cheap\_unit,t} + p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} \\
& + p^{startup\_cost}_{expensive\_unit} \cdot v^{on\_up.}_{expensive\_unit,t} + p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t}
\end{aligned}
$$

subject to:

$$
\begin{aligned}
p^{flow\_profile}_{demand,t} =& v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,out,t} =& p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t}\\
v^{flow}_{expensive\_link,out,t} =& p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t}\\
v^{flow}_{triangle,out,t} =& p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,in,t} =& v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t}\\
v^{flow}_{expensive\_supply,t} =& v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{expensive\_supply,t} =& p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t}\\
v^{flow}_{cheap\_supply,t} =& p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{on}_{cheap\_unit,t} =& v^{on}_{cheap\_unit,t-1} + v^{on\_up}_{cheap\_unit,t} - v^{on\_down}_{cheap\_unit,t}\\
v^{on}_{expensive\_unit,t} =& v^{on}_{expensive\_unit,t-1} + v^{on\_up}_{expensive\_unit,t} - v^{on\_down}_{expensive\_unit,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{on\_up}_{cheap\_unit,t} \leq& 1 - v^{on}_{cheap\_unit,t} - \sum_{t'=1}^{MDT-1} v^{on\_down}_{cheap\_unit,t-t'}\\
v^{on\_down}_{cheap\_unit,t} \leq& v^{on}_{cheap\_unit,t} - \sum_{t'=1}^{MUT-1} v^{on\_up}_{cheap\_unit,t-t'}\\
v^{on\_up}_{expensive\_unit,t} \leq& 1 - v^{on}_{expensive\_unit,t} - \sum_{t'=1}^{MDT-1} v^{on\_down}_{expensive\_unit,t-t'}\\
v^{on\_down}_{expensive\_unit,t} \leq& v^{on}_{expensive\_unit,t} - \sum_{t'=1}^{MUT-1} v^{on\_up}_{expensive\_unit,t-t'}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
0 \leq & v^{flow}_{cheap\_link,in,t} \\
0 \leq & v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq & v^{flow}_{expensive\_link,in,t} \\
0 \leq & v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq & v^{flow}_{triangle,in,t} \\
0 \leq & v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle}
\end{aligned}
$$

$$
\begin{aligned}
0 \leq & v^{flow}_{expensive\_supply,t} && \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \cdot v^{on}_{expensive\_unit,t} \\
0 \leq & v^{flow}_{expensive\_source,t}
\end{aligned}
$$

$$
\begin{aligned}
0 \leq & v^{flow}_{cheap\_supply,t} && \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \cdot v^{on}_{cheap\_unit,t} \\
0 \leq & v^{flow}_{cheap\_source,t}
\end{aligned}
$$

$$
\begin{aligned}
0 \leq & v^{on}_{cheap\_unit,t} && \leq 1 \\
0 \leq & v^{on\_up}_{cheap\_unit,t} && \leq 1 \\
0 \leq & v^{on\_down}_{cheap\_unit,t} && \leq 1 \\
0 \leq & v^{on}_{expensive\_unit,t} && \leq 1 \\
0 \leq & v^{on\_up}_{expensive\_unit,t} && \leq 1 \\
0 \leq & v^{on\_down}_{expensive\_unit,t} && \leq 1
\end{aligned}
$$

1.7 investments = fixed

The goal of this setup is to test basic investments. To that end the resolution = fixed setup is expanded with investment variables. The expensive unit is already invested in to incentivise the model to keep using the expensive unit. In the formulation below, for brevity, the investment cost is to be interpreted as the cost per unit, as opposed to the cost per MW(h).

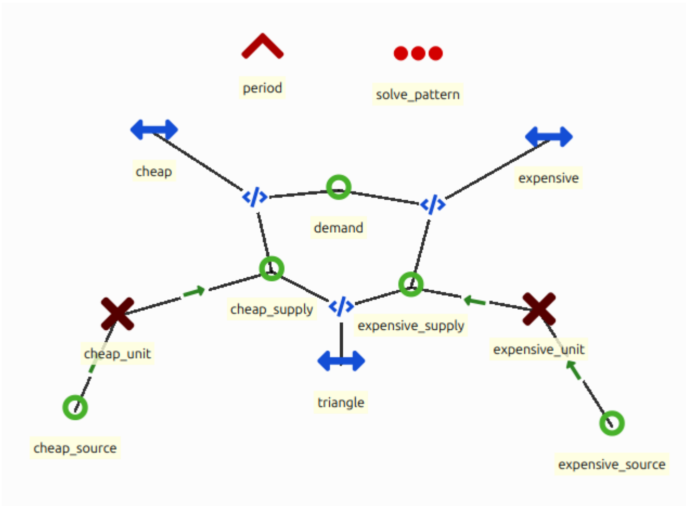

Figure 1.8: Implementation of investment = fixed in the ines spec.

Formulation to obtain the exact solution for this setup:

minimise:

$$
\begin{aligned}
z = & p_{cheap\_unit}^{investment\_cost} \cdot v_{cheap\_unit}^{investment} + p_{expensive\_unit}^{investment\_cost} \cdot (v_{expensive\_unit}^{investment} - p_{expensive\_unit}^{existing\_units}) \\
& + \sum_{t} p_{cheap\_source,t}^{other\_operational\_costs} \cdot v_{cheap\_source,t}^{flow} + p_{expensive\_source,t}^{other\_operational\_costs} \cdot v_{expensive\_source,t}^{flow}
\end{aligned}
$$

subject to:

$$
\begin{aligned}
p^{flow\_profile}_{demand,t} =& v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t}\\
\\
v^{flow}_{cheap\_link,out,t} =& p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t}\\
v^{flow}_{expensive\_link,out,t} =& p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t}\\
v^{flow}_{triangle,out,t} =& p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t}\\
\\
v^{flow}_{cheap\_link,in,t} =& v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t}\\
v^{flow}_{expensive\_supply,t} =& v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t}\\
\\
v^{flow}_{expensive\_supply,t} =& p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t}\\
v^{flow}_{cheap\_supply,t} =& p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t}\\
\\
v^{flow}_{expensive\_supply,t} \leq& p^{capacity}_{expensive\_unit} \cdot v^{investment}_{expensive\_unit}\\
v^{flow}_{cheap\_supply,t} \leq& p^{capacity}_{cheap\_unit} \cdot v^{investment}_{cheap\_unit}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
0 \leq& v^{flow}_{cheap\_link,in,t}\\
0 \leq& v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link}\\
0 \leq& v^{flow}_{expensive\_link,in,t}\\
0 \leq& v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link}\\
0 \leq& v^{flow}_{triangle,in,t}\\
0 \leq& v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle}\\
0 \leq& v^{flow}_{expensive\_supply,t}\\
0 \leq& v^{flow}_{expensive\_source,t}\\
0 \leq& v^{flow}_{cheap\_supply,t}\\
0 \leq& v^{flow}_{cheap\_source,t}\\
0 \leq& v^{investment}_{cheap\_unit}\\
p^{existing\_units}_{expensive\_unit} \leq& v^{investment}_{expensive\_unit}
\end{aligned}
$$

1.8 storage

The goal of this setup is to test an ideal storage formulation. To that end, storage and a renewable unit are added to the resolution = fixed setup. The storage unit is located at the cheap supply side and the renewable is located at the expensive supply side. Storage starts empty.

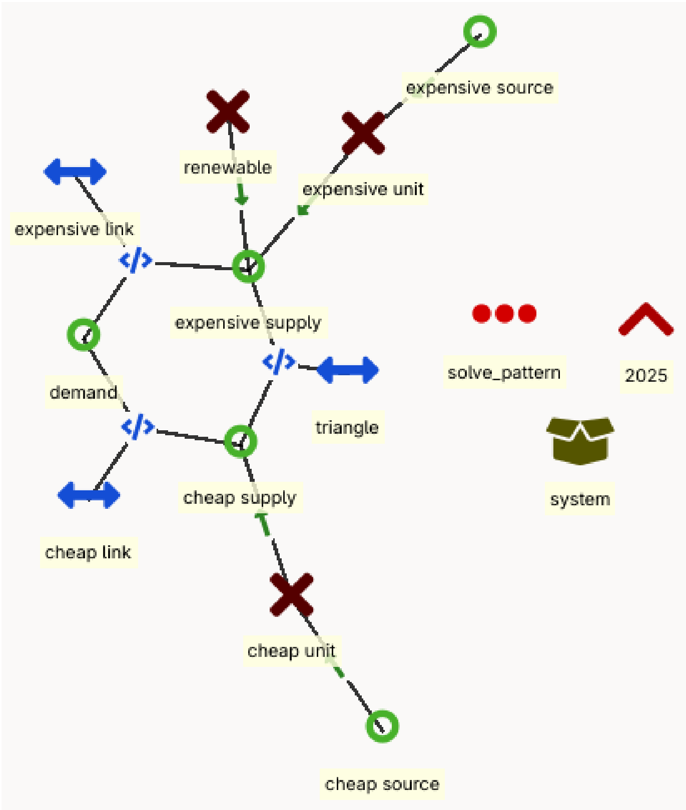

Figure 1.9: Implementation of storage in the ines spec.

Formulation to obtain the exact solution for this setup:

minimise:

$$z = \sum_t p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} + p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t}$$

subject to:

$$
\begin{aligned}
p^{flow\_profile}_{demand,t} =& v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,out,t} =& p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t} \\
v^{flow}_{expensive\_link,out,t} =& p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t} \\
v^{flow}_{triangle,out,t} =& p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,in,t} + v^{charge}_{storage,t} =& v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t} + p^{efficiency,discharge}_{storage} \cdot v^{discharge}_{storage,t} \\
v^{flow}_{expensive\_supply,t} + v^{flow}_{renewable,t} =& v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{expensive\_supply,t} =& p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t} \\
v^{flow}_{cheap\_supply,t} =& p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{state}_{storage,t} =& p^{efficiency}_{storage} \cdot v^{state}_{storage,t-1} + p^{efficiency,charge}_{storage} \cdot v^{charge}_{storage,t} - v^{discharge}_{storage,t} \\
v^{state}_{storage,1} =& p^{efficiency,charge}_{storage} \cdot v^{charge}_{storage,1} - v^{discharge}_{storage,1}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
0 \leq & v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq & v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq & v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle}
\end{aligned}
$$

$$
\begin{aligned}
0 \leq & v^{flow}_{expensive\_supply,t} && \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
0 \leq & v^{flow}_{cheap\_supply,t} && \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit}
\end{aligned}
$$

$$
0 \leq v^{flow}_{renewable,t} \leq p^{flow\_profile}_{renewable,t} \cdot p^{capacity}_{renewable} \cdot p^{existing\_units}_{renewable}
$$

$$
\begin{aligned}
0 \leq & v^{charge}_{storage,t} && \leq p^{capacity\_charge}_{storage} \cdot p^{existing\_units}_{storage} \\
0 \leq & v^{discharge}_{storage,t} && \leq p^{capacity\_discharge}_{storage} \cdot p^{existing\_units}_{storage} \\
0 \leq & v^{state}_{storage,t} && \leq p^{capacity}_{storage} \cdot p^{existing\_units}_{storage}
\end{aligned}
$$

1.9 rolling horizon

The goal of the setup is to test a rolling horizon. To that end the investment = fixed setup is expanded with a second horizon. Each horizon uses the same formulation. In the first horizon, there is already a small investment in expensive units and the model needs to invest in cheaper units. The second window has a higher demand requiring additional investments but there is also a shift in commodity prices such that the more 'expensive' unit becomes more attractive, yet there are already the investments from the first window in the 'cheaper' units.

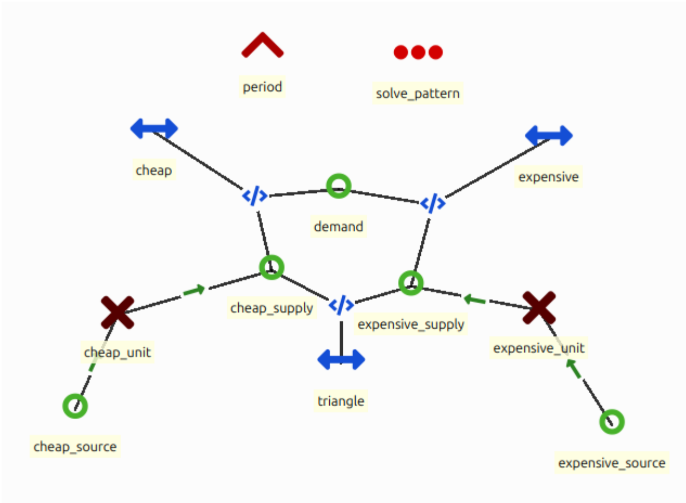

Figure 1.10: Implementation of unit commitment in the ines spec.

Formulation to obtain the exact solution for this setup:

minimise:

$$
\begin{aligned}
z = & p^{investment\_cost}_{cheap\_unit} \cdot (v^{investment}_{cheap\_unit} - p^{existing\_units}_{cheap\_unit}) + p^{investment\_cost}_{expensive\_unit} \cdot (v^{investment}_{expensive\_unit} - p^{existing\_units}_{expensive\_unit}) \\
& + \sum_t p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} + p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t}
\end{aligned}
$$

subject to:

$$
\begin{aligned}
p^{flow\_profile}_{demand,t} =& v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,out,t} =& p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t}\\
v^{flow}_{expensive\_link,out,t} =& p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t}\\
v^{flow}_{triangle,out,t} =& p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,in,t} =& v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t}\\
v^{flow}_{expensive\_supply,t} =& v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{expensive\_supply,t} =& p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t}\\
v^{flow}_{cheap\_supply,t} =& p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{expensive\_supply,t} \leq& p^{capacity}_{expensive\_unit} \cdot v^{investment}_{expensive\_unit}\\
v^{flow}_{cheap\_supply,t} \leq& p^{capacity}_{cheap\_unit} \cdot v^{investment}_{cheap\_unit}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
0 \leq& v^{flow}_{cheap\_link,in,t}\\
0 \leq& v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link}\\
0 \leq& v^{flow}_{expensive\_link,in,t}\\
0 \leq& v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link}\\
0 \leq& v^{flow}_{triangle,in,t}\\
0 \leq& v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle}\\
0 \leq& v^{flow}_{expensive\_supply,t}\\
0 \leq& v^{flow}_{expensive\_source,t}\\
0 \leq& v^{flow}_{cheap\_supply,t}\\
0 \leq& v^{flow}_{cheap\_source,t}\\
p^{existing\_units}_{cheap\_unit} \leq& v^{investment}_{cheap\_unit}\\
p^{existing\_units}_{expensive\_unit} \leq& v^{investment}_{expensive\_unit}
\end{aligned}
$$

1.10 combined system

The goal of this setup is to test systems that combine some of the different features. This setup is a combination of all the previous setups (except for the stochastics and rolling horizon to reduce the complexity for calculations by hand). It is sector coupled for the flexible time steps. The gas sector consists of import, demand and gas provision to a gas plant. The electricity sector consists of the gas plant, a nuclear plant, batteries, onshore PV, offshore wind, electricity import and demand. The demand is in the distribution grid, the import and offshore wind is in a separate node. The nuclear plant, batteries and import are already there, but the model can invest in a limited amount of gas plants and renewable energy systems.

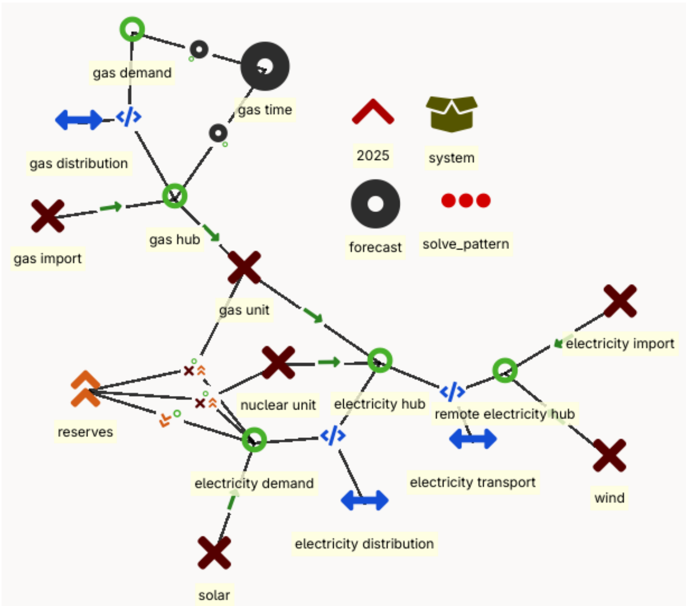

Figure 1.11: combined system

Formulation to obtain the exact solution for this setup:

minimise:

$$
\begin{aligned}
z = & \sum_{t \in t_{gas}} p^{other\_operational\_costs}_{gas\_import,t} \cdot v^{flow}_{gas\_import,t} \cdot \Delta_{t_{gas}} + \sum_{t} p^{other\_operational\_costs}_{electricity\_import,t} \cdot v^{flow}_{electricity\_import,t} \\
& + \sum_{t} p^{startup\_cost}_{nuclear\_unit} \cdot v^{on\_up.}_{nuclear\_unit,t} + p^{other\_operational\_costs}_{nuclear\_unit} \cdot v^{flow}_{nuclear\_unit,t} \\
& + \sum_{t} p^{penalty}_{demand} \cdot v^{slack}_{demand,t} + p^{procurement\_cost}_{nuclear\_unit,t} \cdot v^{flow,reserve}_{nuclear\_unit,t} + p^{procurement\_cost}_{gas\_unit,t} \cdot v^{flow,reserve}_{gas\_unit,t} \\
& + p^{investment\_cost}_{gas\_unit} \cdot (v^{investment}_{gas\_unit} - p^{existing\_units}_{gas\_unit}) \\
& + p^{investment\_cost}_{wind\_unit} \cdot (v^{investment}_{wind\_unit} - p^{existing\_units}_{wind\_unit}) + p^{investment\_cost}_{solar\_unit} \cdot (v^{investment}_{solar\_unit} - p^{existing\_units}_{solar\_unit})
\end{aligned}
$$

subject to:

$$p^{reserve}_{electricity\_demand,t} =< v^{flow,reserve}_{gas\_unit,t} + v^{flow,reserve}_{nuclear\_unit,t} + v^{slack}_{electricity\_demand,t}$$

$$\begin{aligned}
v^{flow}_{gas_unit,out,t} + v^{flow,reserve}_{gas_unit,t} &\leq p^{capacity}_{gas_unit} \cdot v^{investment}_{gas_unit} \\
v^{flow}_{nuclear_unit,t} + v^{flow,reserve}_{nuclear_unit,t} &\leq p^{capacity}_{nuclear_unit} \cdot p^{existing_units}_{nuclear_unit} \cdot v^{on}_{nuclear_unit,t}
\end{aligned}$$

$$\begin{aligned}
p^{flow_profile}_{gas_demand,t} &= v^{flow}_{gas_distribution,out,t} \\
v^{flow}_{gas_import,t} &= v^{flow}_{gas_distribution,in,t} + v^{flow}_{gas_unit,in,t} \\
p^{flow_profile}_{electricity_demand,t} &= v^{flow}_{electricity_distribution,out,t} + v^{flow}_{solar_unit,t} \\
v^{flow}_{electricity_distribution,in,t} &= v^{flow}_{electricity_transport,out,t} + v^{flow}_{nuclear_unit,t} + v^{flow}_{gas_unit,out,t} \\
&\quad - v^{charge}_{battery,t} + p^{efficiency,discharge}_{battery} \cdot v^{discharge}_{battery,t} \\
v^{flow}_{electricity_transport,in,t} &= v^{flow}_{wind_unit,t} + v^{flow}_{electricity_import,t}
\end{aligned}$$

$$\begin{aligned}
v^{flow}_{gas_distribution,out,t} &= p^{efficiency}_{gas_distribution} \cdot v^{flow}_{gas_distribution,in,t} \\
v^{flow}_{electricity_distribution,out,t} &= p^{efficiency}_{electricity_distribution} \cdot v^{flow}_{electricity_distribution,in,t} \\
v^{flow}_{electricity_transport,out,t} &= p^{efficiency}_{electricity_transport} \cdot v^{flow}_{electricity_transport,in,t}
\end{aligned}$$

$$\sum_{t \in t_{gas}} v^{flow}_{gas\_unit,out,t} \cdot 1h = p^{efficiency}_{gas\_unit} \cdot v^{flow}_{gas\_unit,in,t_{gas}} \cdot \Delta_{t_{gas}}$$

$$\begin{aligned}
v^{on}_{nuclear_unit,t} &= v^{on}_{nuclear_unit,t-1} + v^{on_up}_{nuclear_unit,t} - v^{on_down}_{nuclear_unit,t} \\
v^{on_up}_{nuclear_unit,t} &\leq 1 - v^{on}_{nuclear_unit,t} - \sum_{t'=1}^{MDT-1} v^{on_down}_{nuclear_unit,t-t'} \\
v^{on_down}_{nuclear_unit,t} &\leq v^{on}_{nuclear_unit,t} - \sum_{t'=1}^{MUT-1} v^{on_up}_{nuclear_unit,t-t'}
\end{aligned}$$

$$\begin{aligned}
v^{state}_{storage,t} &= p^{efficiency}_{storage} \cdot v^{state}_{storage,t-1} + p^{efficiency,charge}_{storage} \cdot v^{charge}_{storage,t} - v^{discharge}_{storage,t} \\
v^{state}_{storage,1} &= p^{efficiency,charge}_{storage} \cdot v^{charge}_{storage,1} - v^{discharge}_{storage,1}
\end{aligned}$$

bound to:

$$
\begin{aligned}
0 \leq\ & v^{slack}_{electricity\_demand,t} \\
0 \leq\ & v^{flow,reserve}_{gas\_unit,t} && \leq p^{max\_reserve}_{gas\_unit} \cdot v^{investment}_{gas\_unit} \\
0 \leq\ & v^{flow,reserve}_{nuclear\_unit,t} && \leq p^{max\_reserve}_{nuclear\_unit} \cdot p^{existing\_units}_{nuclear\_unit} \\
\\
0 \leq\ & v^{flow}_{gas\_distribution,out,t} && \leq p^{capacity}_{gas\_distribution} \cdot p^{existing\_units}_{gas\_distribution} \\
0 \leq\ & v^{flow}_{electricity\_distribution,out,t} && \leq p^{capacity}_{electricity\_distribution} \cdot p^{existing\_units}_{electricity\_distribution} \\
0 \leq\ & v^{flow}_{electricity\_transport,out,t} && \leq p^{capacity}_{electricity\_transport} \cdot p^{existing\_units}_{electricity\_transport} \\
\\
0 \leq\ & v^{flow}_{gas\_import,t} && \leq p^{capacity}_{gas\_import} \cdot p^{existing\_units}_{gas\_import} \\
0 \leq\ & v^{flow}_{electricity\_import,t} && \leq p^{capacity}_{electricity\_import} \cdot p^{existing\_units}_{electricity\_import} \\
\\
0 \leq\ & v^{flow}_{nuclear\_unit,t} && \leq p^{capacity}_{nuclear\_unit} \cdot p^{existing\_units}_{nuclear\_unit} \\
0 \leq\ & v^{on}_{nuclear\_unit,t} && \leq 1 \\
0 \leq\ & v^{on\_up}_{nuclear\_unit,t} && \leq 1 \\
0 \leq\ & v^{on\_down}_{nuclear\_unit,t} && \leq 1 \\
\\
0 \leq\ & v^{flow}_{gas\_unit,in,t} && \leq \frac{1}{p^{efficiency}_{gas\_unit}} \cdot p^{capacity}_{gas\_unit} \cdot v^{investment}_{gas\_unit} \\
0 \leq\ & v^{flow}_{gas\_unit,out,t} && \leq p^{capacity}_{gas\_unit} \cdot v^{investment}_{gas\_unit} \\
\\
0 \leq\ & v^{flow}_{solar\_unit,t} && \leq p^{flow\_profile}_{solar\_unit,t} \cdot p^{capacity}_{solar\_unit} \cdot v^{investment}_{solar\_unit} \\
0 \leq\ & v^{flow}_{wind\_unit,t} && \leq p^{flow\_profile}_{wind\_unit,t} \cdot p^{capacity}_{wind\_unit} \cdot v^{investment}_{wind\_unit} \\
\\
0 \leq\ & v^{charge}_{battery,t} && \leq p^{capacity\_charge}_{battery} \cdot p^{existing\_units}_{battery} \\
0 \leq\ & v^{discharge}_{battery,t} && \leq p^{capacity\_discharge}_{storage} \cdot p^{existing\_units}_{battery} \\
0 \leq\ & v^{state}_{battery,t} && \leq p^{capacity}_{storage} \cdot p^{existing\_units}_{battery} \\
\\
p^{existing\_units}_{gas\_unit} \leq\ & v^{investment}_{gas\_unit} && \leq p^{max\_cumulative}_{gas\_unit} \\
p^{existing\_units}_{solar\_unit} \leq\ & v^{investment}_{solar\_unit} && \leq p^{max\_cumulative}_{solar\_unit} \\
p^{existing\_units}_{wind\_unit} \leq\ & v^{investment}_{wind\_unit} && \leq p^{max\_cumulative}_{wind\_unit}
\end{aligned}
$$

Appendix A

Extended Certification process

The certification setups from Chapter 1 are selected for a proof of concept. The full certification process consists of:

equation tests are a systematic group of tests to test each equation of energy system models individually, starting from a simple energy balance on one node with 1 demand unit and on supply unit

method tests are tests that cover multiple equations to show off a certain feature (various examples can be found in Chapter 1)

system tests are more comprehensible, similar to IEEE networks (an example can be found in the final setup of Chapter 1)

For the equation tests, inspiration can be taken from the different formulation in the proof of concept of the certification process. For the method tests, table A is an indicative expansion of the certification process. The system tests are a point of discussion for the consortium of the ines spec.

Table A.1: Expansion of the certification process

name	description
network = copper plate	Energy balance where the demand equals the supply for a single time step. Two units where one is cheaper than the other.
network = radial	Two supply units, each connected with a directional line to the demand. The cheaper unit has the direction from the demand to the supply so the setup has no choice but to choose the more expensive unit.

network = meshed	Energy balance where the demand equals the supply for a single time step. Triangular network with the two supply units and demand unit. The cheapest unit should be chosen.
resolution = fixed	Same setup but with a time series for all units with the same resolution.
resolution = variable	Same setup but the time series has a variable resolution.
resolution = flexible	Same setup but the two units have different resolutions (1h for expensive unit and 2h for cheap unit).
stochastic	Meshed network, fixed resolution, expensive and cheap unit. The units have the same stochastics.
stochastic = flexible	Same setup but the units have different stochastics.
CO2 cap	Meshed network, fixed resolution, expensive and cheap unit. The cheap unit produces CO2 and is limited by the cap.
reserves	Meshed network, fixed resolution, expensive and cheap unit. The expensive unit is able to provide reserves but not enough to entirely avoid load shedding.
cogeneration	Meshed network, fixed resolution, two supply units and two demand units but one one supply is cogeneration and one demand is for heat. The setup is heat driven as only the electricity grid has two supply units.
ramping	Meshed network, fixed resolution, expensive and cheap unit. The cheap unit is not able to ramp. The demand profile is either an elevated sine or piecewise linear _/.
flow = fixed	Meshed network, fixed resolution, expensive and cheap unit. The expensive unit is forced to produce a certain flow.

flow = limited	Meshed network, fixed resolution, expensive and cheap unit. The expensive unit is forced to produce within certain limits. This test also represents curtailment.
storage = intra	Meshed network, fixed resolution, one storage unit and one supply unit. The supply unit has a fixed output that does not correspond to the demand. The storage has to be used to be feasible.
storage = inter	Same setup but there are multiple periods.
part load efficiency	Meshed network, fixed resolution, expensive and cheap unit. The partload efficiency of the cheap unit makes it that the overall cost is higher than the expensive unit.
MUT	Meshed network, fixed resolution, expensive storage and cheap unit. By setting the minimum production equal to the maximal production the output of the unit becomes fixed. By setting the minimum up time (MUT) equal to the lifetime of the unit, whenever the unit delivers an output, that output will need to be maintained for the remainder of the simulation. That means that there will appear horizontal lines for each level of required output. To make sure that the system works, a storage unit is used to get rid of the excess power. The storage unit is very expensive such that it would be avoided if it would not be needed.
MDT	Same setup, but there is a moment where the demand is zero. Normally, the plant would shut down, but due to the MDT it needs to remain on.
economic representation	Meshed network, fixed resolution, expensive and cheap unit. The numbers will be different from other setups due to the economic representation.

investment = fixed	Meshed network, fixed resolution, expensive and cheap unit. Expensive unit is already invested in.
investment = cumulative	Same setup but a cumulative cap
investment = multi year	Same setup but multiple periods.
retirement = fixed	Same setup Expensive unit is already invested in but is sunset in the second period.
retirement = economic	Same setup but the retirement is not fixed.
representative periods	
blended representative periods	
rolling horizon	Same setup but the periods are separated.
benders decomposition	Same setup but the investment and operation are separated from each other.
mini Europe	A miniature version of europe with very limited temporal aspects (a day = 6 hours, a season is 1 day). Each country has a demand but only its most prominent technologies. Complex setup with storage, cogeneration, CO2 cap, ramping, flow limits, part load efficiency, MUT/MDT.